

100 Days of Azure DevOps

Day 21 Handout — A robot that runs the build so you can stop being the robot

3 - Continuous Integration · Personal lab only · Educational content · Not a sales pitch

Architecture A — pipeline run graph (YAML file → logs)

Piece	YAML / product	What it does
Pipeline definition	azure-pipelines.yml in Git	Versioned with the code
Trigger	trigger: branches include main	Start a run on push
Pool	pool.vmlImage: ubuntu-latest	Microsoft-hosted agent VM
Job	jobs.job (implicit job if only steps)	Unit of agent allocation
Step	script / task / checkout	Process on the agent
Run / logs	Pipelines → Runs → Job log	The product you actually debug
Parallel job SKU	Org settings → Parallel jobs	Without it, runs queue forever

Architecture B — YAML vs Classic (decide once for this series)

Need	Default	Classic only if
Reviewable CI	YAML in /pipelines or repo root	You are maintaining a legacy GUI release
Branch diffs	git blame the YAML	Classic JSON in ADO, not in Git
Reuse	templates: later days	Task groups in Classic
PR validation	pr: + branch policy build validation	Classic build attached as policy
This 100-day lab	YAML only	Do not create a Classic build 'to compare' beyond a look
Disable Classic	Org/project: Disable creation of classic pipelines (if present)	Admin toggle — optional discipline

Architecture C — pitfalls that never start a run

Pitfall	What you see	Fix
YAML not in default branch	Pipeline exists, push to main does nothing	Merge YAML to main; trigger includes main
Wrong repo	Pipeline bound to day05-overview	Create pipeline on azure-100-labs
No hosted parallel job	Waiting for an available agent	Org settings → Parallel jobs; free Microsoft-hosted
pool: Default by copy-paste	Job queued on empty self-hosted pool	pool: vmlImage: ubuntu-latest
Classic editor curiosity	GUI pipeline with no Git history	Delete it; keep YAML
Checkout skipped + script needs files	cat: No such file	steps include implicit checkout unless checkout: none

One-liner to remember

Pipelines are robots that run your build so you can stop being the robot.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal project azure-100-labs. Microsoft-hosted ubuntu-latest only. No Classic definition as the source of truth.

- 1 Pipelines → Create Pipeline → Azure Repos Git → azure-100-labs → Starter pipeline.
- 2 Replace the body with the YAML in this handout. Save as /azure-pipelines.yml (repo root) on a feature branch or main via PR.
- 3 Run pipeline. Open the job log. Find the image / Agent name / Agent.OS lines — that is hosted proof.
- 4 Confirm YAML has pool: vmImage: ubuntu-latest (not pool: Default).
- 5 Org settings → Agent pools / Parallel jobs. Note Microsoft-hosted vs any Default pool. Do not register an agent today.
- 6 Optional: Project settings → Pipelines → Settings → disable creation of classic build/release if the toggle exists.
- 7 PR the YAML if main is policy-protected. Keep the first green run ID in notes.

azure-pipelines.yml — hello on Microsoft-hosted ubuntu-latest

```
trigger:
  branches:
    include:
      - main

pool:
  vmImage: ubuntu-latest

steps:
  - checkout: self
  - script: |
      echo Hello from Day 21
      echo Agent.OS=$(Agent.OS)
      echo Agent.Name=$(Agent.Name)
      echo Build.SourceVersion=$(Build.SourceVersion)
    displayName: Hello and prove the agent
```

Done checklist

- Can name trigger, pool, job, step, run log
- Green run on ubuntu-latest with Hello in the log
- YAML lives in Git (not only Classic GUI)
- Did not register a self-hosted agent

Tomorrow — Day 22

Microsoft-hosted vs self-hosted agents